

Tech & Teach

Digital Chip Design Program

RTL Design

Task #04

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Structural modeling & testbench

Version 1.0

Task #1.1

Project Summary x Schematic x top_module.sv x **comparator.sv** x or2.sv x and2.v x not1.sv x

C:/top_module/top_module.srscs/sources_1/new/comparator.sv

Q [Icons]

```
1 module comparator (AeqB, AltB, AgtB, a, b);
2
3     input [3:0] a, b;
4     output AeqB, AltB, AgtB;
5
6     assign AeqB = (a == b) ? 1 : 0;
7     assign AltB = (a < b) ? 1 : 0;
8     assign AgtB = (a > b) ? 1 : 0;
9
10 endmodule
```

Project Summary x Schematic x top_module.sv x comparator.sv x **or2.sv** x and2.v x not1.sv x

C:/top_module/top_module.srscs/sources_1/new/or2.sv

Q [Icons]

```
1 module or2(input logic A, input logic B, output logic Y);
2     assign Y = A | B;
3 endmodule
```

Project Summary x Schematic x top_module.sv x comparator.sv x or2.sv x **and2.v** x not1.sv x

C:/top_module/top_module.srscs/sources_1/new/and2.v

Q [Icons]

```
1 module and2(input A, input B, output Y);
2     assign Y = A & B;
3 endmodule
```

Project Summary x Schematic x top_module.sv x comparator.sv x or2.sv x and2.v x **not1.sv** x

C:/top_module/top_module.srscs/sources_1/new/not1.sv

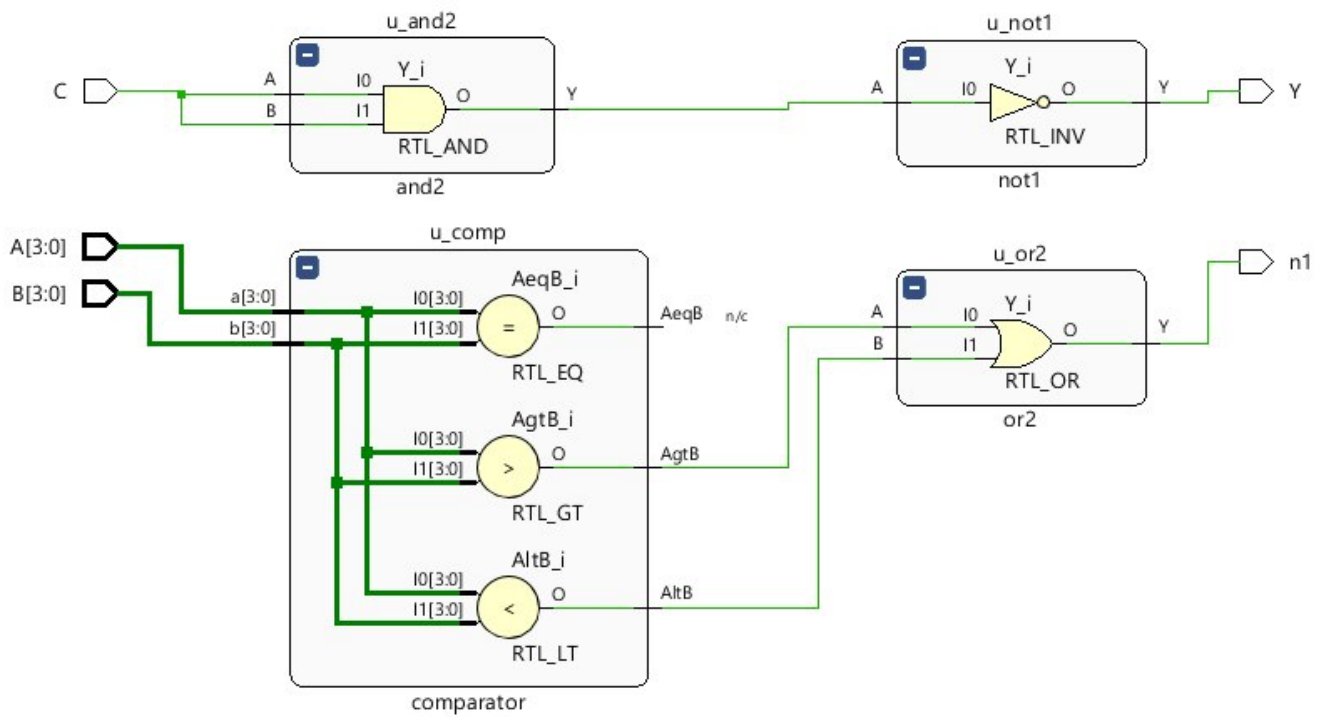
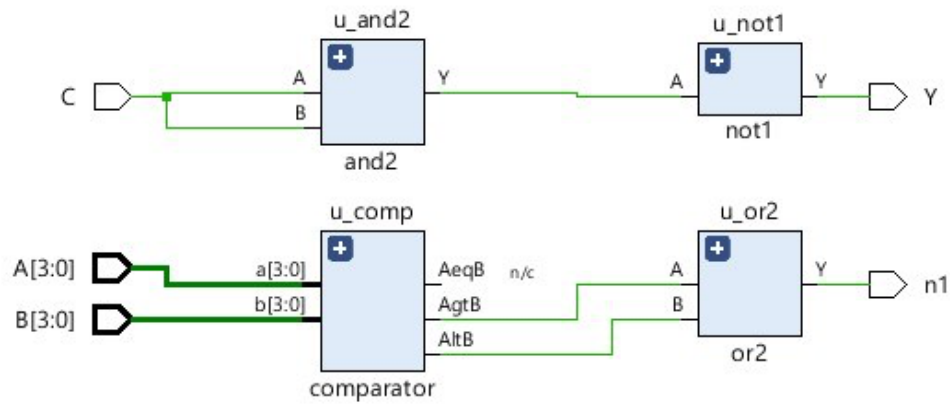
Q [Icons]

```
1 module not1(input logic A, output logic Y);
2     assign Y = ~A;
3 endmodule
```

```

22 module top_module (
23     input logic [3:0] A,
24     input logic [3:0] B,
25     input logic C,
26     output logic Y,
27     output logic n1
28 );
29
30     logic AgtB, AltB;
31     logic n2;
32
33     comparator u_comp (
34         .AeqB(),
35         .AltB(AltB),
36         .AgtB(AgtB),
37         .a(A),
38         .b(B)
39     );
40
41     or2 u_or2 (
42         .A(AgtB),
43         .B(AltB),
44         .Y(n1)
45     );
46
47     and2 u_and2 (
48         .A(C),
49         .B(C),
50         .Y(n2)
51     );
52
53     not1 u_not1 (
54         .A(n2),
55         .Y(Y)
56     );
57
58 endmodule

```



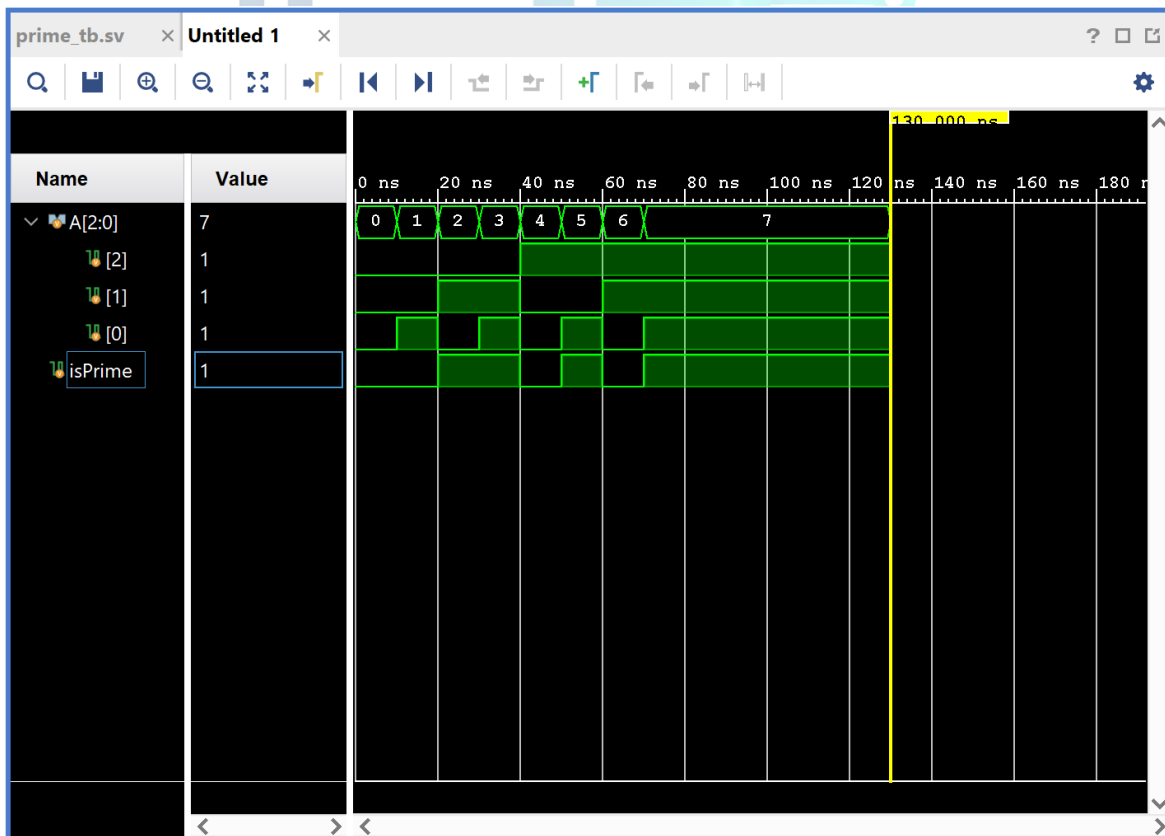
Task #1.2

```

module prime_tb;
    logic [2:0] A;
    logic isPrime;
    prime uut (
        .A(A), .isPrime(isPrime) );
    initial begin
        $display("A2 A1 A0 | isPrime");
        $display("-----");
        for (int i = 0; i < 8; i++) begin
            A = i;
            #10;
            $display(" %b %b %b | %b", A[2], A[1], A[0],
        end
        $finish;
    end
endmodule

```

A2	A1	A0		isPrime
0	0	0		0
0	0	1		0
0	1	0		1
0	1	1		1
1	0	0		0
1	0	1		1
1	1	0		0
1	1	1		1

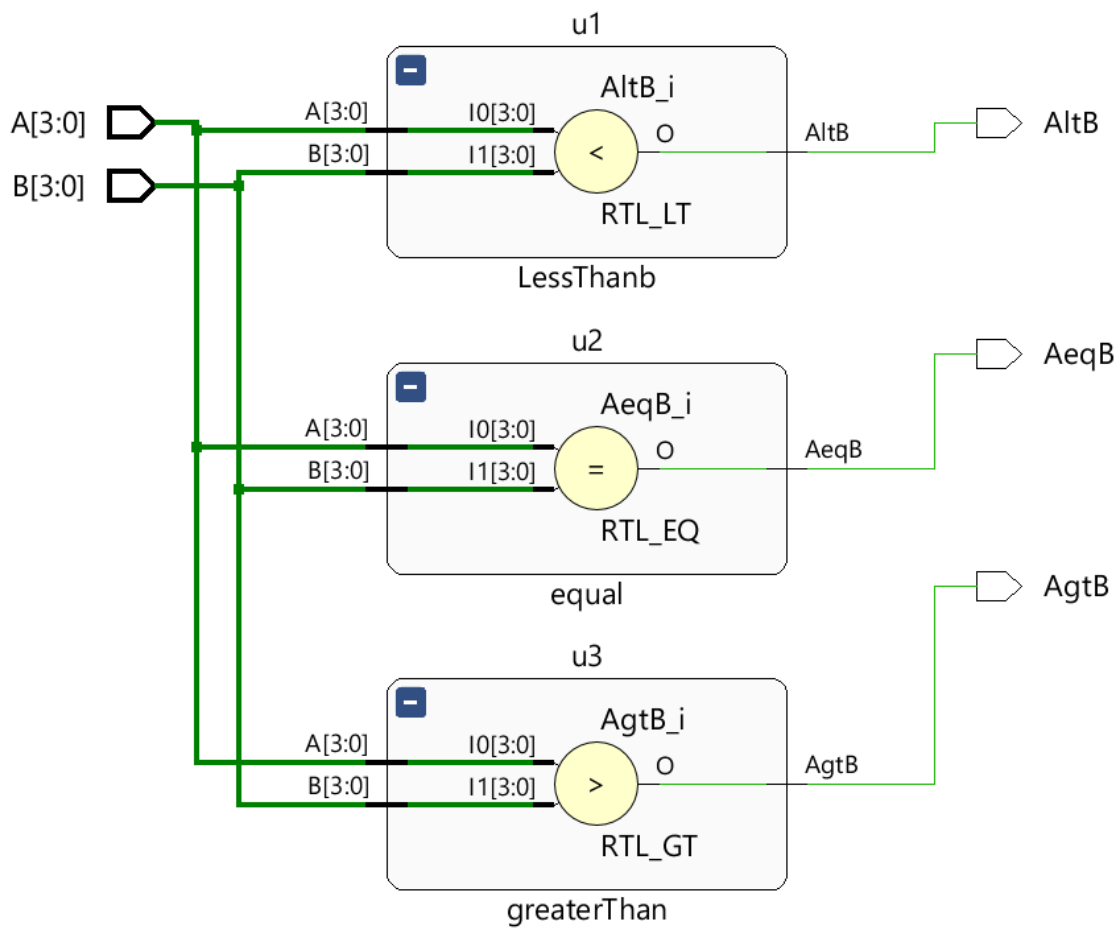
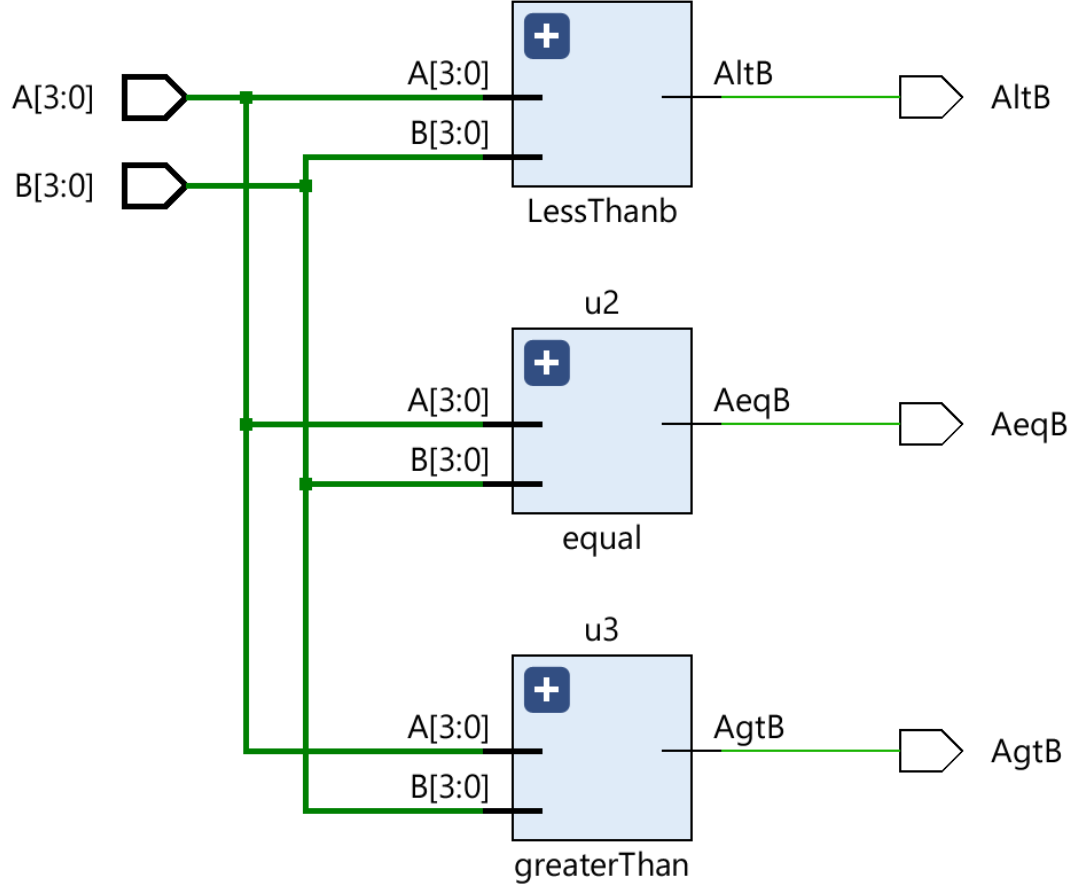


Task #1.3

```
1  module comparator(  
2      input  [3:0] A,  
3      input  [3:0] B,  
4      output AeqB,  
5      output AltB,  
6      output AgtB  
7  );  
8  
9  LessThanb    u1 (AltB, A, B);  
10 equal        u2 (AeqB, A, B);  
11 greaterThan u3 (AgtB, A, B);  
12  
13 endmodule
```



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```
`timescale 1ns / 1ps
```

```
module comparator_tb();
```

```
logic [3:0] A;
```

```
logic [3:0] B;
```

```
logic AeqB;
```

```
logic AltB;
```

```
logic AgtB;
```

```
comparator dut(
```

```
    .A(A),
```

```
    .B(B),
```

```
    .AeqB(AeqB),
```

```
    .AltB(AltB),
```

```
    .AgtB(AgtB)
```

```
);
```

```
initial begin
```

```
    for (int i = 0; i < 5'b10000; i++) begin
```

```
        for (int j = 0; j < 5'b10000; j++) begin
```

```
            A = i;
```

```
            B = j;
```

```
            #10;
```

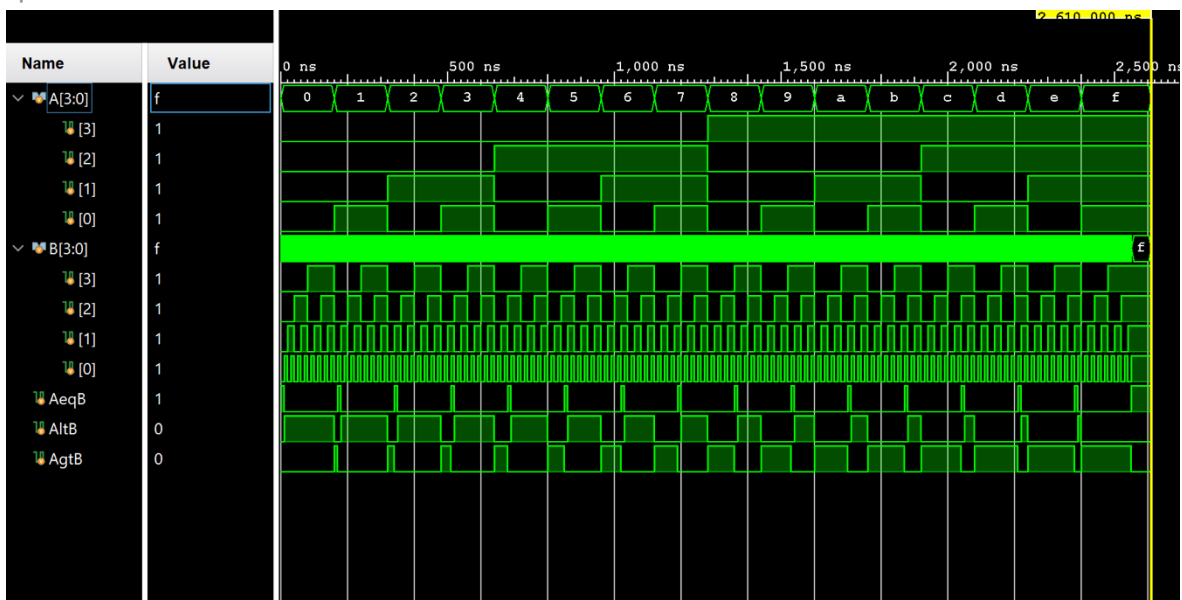
```
        end
```

```
    end
```

```
    #50 $finish;
```

```
end
```

```
endmodule
```



Work distribution

Section	Farah&Joud	Tala&Horiah	Rana&Hoor&Sandra
Report	✓	✓	✓
Task 1.1		✓	
Task 1.2			✓
Task 1.3	✓		